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Multi-Agent AI System Based on LangGraph for 5G/6G Applications

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Abstract

In modern 5G networks, joint control of handover decisions and mobile edge computing (MEC) offloading requires coordination across multiple timescales and competing objectives that reactive rule-based algorithms cannot adequately address. This project proposes a hierarchical multi-agent system that divides network control by timescale. At the lower level, each gNB has a local tower agent that makes fast handover and MEC placement decisions using the current UE intent vector. A deterministic cross-layer supervisor resolves conflicts requiring cross-tower visibility, while a strategic optimizer running every 60 seconds selects from eight pre-validated policy bundles using a large language model (LLM). The LLM acts only as a bounded advisor, while a deterministic safety shield enforces hard constraints, rejects unsafe choices, and rolls back harmful changes.

The system is evaluated in a 3GPP-inspired digital twin across four scenarios (3–5 gNBs, 15–45 UEs). The evaluation considers three 5G service classes with competing requirements: URLLC traffic prioritizes ultra-low latency and reliability, eMBB traffic emphasizes high compute throughput and task completion, and mMTC traffic focuses on energy efficiency and stable large-scale connectivity. An ablation study progressing from a 3GPP A3 baseline (B0) through intent-aware tower agents (B1), a deterministic supervisor (B2-noLLM), and guarded LLM bundle selection (B2) shows that the deterministic layers provide most of the performance gains.

B1 reduces URLLC drop by 1.78–1.94 percentage points over B0, while B2-noLLM adds a further 1.02–1.27 percentage points together with substantial eMBB goodput improvements. The LLM layer then provides additional consistent gains across all evaluated scenarios. Two comparable backends are evaluated: GPT-4o-mini and TeleLLM (LLaMA-3-8B-Tele-it). TeleLLM achieves the best URLLC drop and mMTC energy performance, while GPT-4o-mini leads on eMBB goodput. The safety shield modifies or reverses 40–62 percent of LLM selections, yet both B2 modes consistently outperform the deterministic baseline. Across the full ablation, the complete three-layer system reduces URLLC drop by 73 percent relative to the A3 baseline.

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List of Symbols and Abbreviations

3GPP	3rd Generation Partnership Project
5G	Fifth-generation cellular networks
6G	Sixth-generation cellular networks
A3	3GPP A3 handover event
B0	Baseline mode
B1	Intent-aware mode
B2	Guarded LLM mode
B2-noLLM	B2 mode without LLM involvement
DRL	Deep Reinforcement Learning
eMBB	enhanced Mobile Broadband
GFLOPS/s	Giga Floating-Point Operations Per Second
gNB	next-generation base station
KPI	Key Performance Indicator
LLM	Large Language Model
LOS	Line-of-Sight
MARL	Multi-Agent Reinforcement Learning
MEC	Mobile Edge Computing
mMTC	massive Machine-Type Communications
NLOS	Non-Line-of-Sight
P99	99th percentile latency
RAN	Radio Access Network
RSRP	Reference Signal Received Power
SINR	Signal-to-Interference-plus-Noise Ratio
TTT	Time-to-Trigger
UE	User Equipment
UMa	Urban Macro
URLLC	Ultra-Reliable Low-Latency Communications
Xn	Inter-gNB interface

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Chapter 1

Introduction

1.1 Background and Motivation

The deployment of fifth-generation (5G) cellular networks has introduced a much higher level of complexity into wireless network management. Unlike earlier generations, 5G must simultaneously support three fundamentally different service classes on a shared physical infrastructure: Ultra-Reliable Low-Latency Communications (URLLC), enhanced Mobile Broadband (eMBB), and massive Machine-Type Communications (mMTC). [1] These service classes impose competing requirements on latency, throughput, and energy efficiency, creating a multi-objective optimization problem that traditional management approaches cannot handle effectively.

At the physical layer, 5G networks are organized around next-generation base stations (gNBs) serving User Equipment (UE) over the radio access network (RAN). Each gNB may also host a Mobile Edge Computing (MEC) server that executes computation tasks locally, reducing latency and backhaul load compared with centralized cloud execution. [1] When a UE moves between cells, the network performs a handover procedure that transfers radio connectivity from one gNB to another. However, handover also affects ongoing MEC tasks, since tasks running on the source gNB may need to be migrated to the target gNB, introducing additional latency, energy consumption, and possible deadline violations.

This interaction between handover control and MEC placement is the central coordination challenge addressed in this project. A handover decision changes radio conditions while simultaneously affecting task migration cost and edge resource availability. Optimizing one domain independently can therefore degrade the other. A handover controller that ignores ongoing MEC tasks may disrupt computation, while a MEC scheduler that ignores radio dynamics may select inefficient task placement. Effective network control therefore requires coordinated decision-making across both domains and across multiple timescales.

1.2 Problem Statement

The network management problem addressed in this project can be stated as follows: given a 5G network with multiple gNBs, each serving multiple heterogeneous UEs across three service slices, design a control system that makes handover decisions and MEC offloading decisions in a coordinated way, so that radio connectivity and task placement are optimized together rather than independently, in order to minimize URLLC deadline violations, maximize eMBB compute throughput, minimize mMTC device energy consumption, and suppress unnecessary handovers, all in a manner that is safe, interpretable, and scalable across network topologies and load conditions.

This problem is challenging for three reasons that align with the limitations of the three main classes of prior approaches. First, it is a multi-objective problem with competing service requirements, making it impossible to define a single scalar reward or objective. Second, it involves multiple timescales: radio events occur on a millisecond timescale while strategic reconfigurations are appropriate only on a timescale of tens of seconds, because frequent large-scale policy changes introduce instability. Third, it involves cross-tower dependencies that are invisible to any single gNB: a handover decision at one tower creates a task migration that burdens a neighbor tower, and this cross-tower coupling can only be resolved by a controller with global visibility.

1.3 Existing Approaches and Their Limitations

The coordination problem described above has been approached in three main ways in prior work, each with significant limitations.

Mathematical programming and game-theoretic formulations can produce optimal solutions under their stated assumptions, but they tend to rely on simplified models and do not scale well when traffic is dynamic and heterogeneous [2]. They typically cannot accommodate the stochastic variability of real 5G traffic or the non-stationarity introduced by UE mobility.

Multi-Agent Reinforcement Learning (MARL) can learn adaptive policies through interaction with the environment. However, MARL requires extensive offline training, becomes brittle under distribution shift when the topology or traffic pattern changes,

produces opaque decisions that operators cannot inspect, and can become unstable when the reward function or action space is not carefully designed [3], [4]. Deploying a MARL system in a live 5G network requires retraining whenever the environment changes, which is operationally expensive.

More recently, Large Language Models (LLMs) have been applied to network intent translation [5], configuration generation [6], and resource management guidance [7]. Their ability to reason in unfamiliar domains without task-specific training is attractive. However, existing LLM-based methods either require the model to directly output network parameters, introducing unacceptable latency and safety risks, or restrict it to natural-language guidance without closed-loop integration. Neither approach provides the bounded, auditable control that safety-critical infrastructure demands.

1.4 Proposed Solution

This project proposes a fundamentally different approach. Rather than optimizing by function, the control problem is decomposed by timescale. Fast, deterministic agents handle all millisecond-scale operations: handover triggering, MEC task admission and placement, and cross-layer conflict resolution. A slow LLM-based optimizer, operating on a 60-second cycle, selects among eight pre-validated policy bundles that bias the behavior of the fast agents. This timescale separation ensures that the LLM's inference latency, which ranges from a few seconds to tens of seconds depending on the deployment infrastructure, has no impact on radio-layer safety.

The key architectural insight is that, in safety-critical infrastructure, the appropriate role for an LLM is not fine-grained parameter control but bounded strategic judgment. The LLM selects a named operating regime from a finite, pre-validated library. A deterministic safety shield then validates, and if necessary, overrides the LLM's selection before it reaches the network. This design means that the network is never exposed to arbitrary LLM output: the worst-case outcome of an LLM failure is that the shield reverts to the previous safe bundle.

1.5 Ablation Design and Contribution Isolation

A staged ablation methodology is employed to isolate the contribution of each architectural layer. The system is evaluated in four configurations:

1. B0 is a standard 3GPP A3 handover baseline with greedy MEC placement, representing uncoordinated operation.
2. B1 adds intent-aware tower agents that adapt handover and MEC parameters to per-UE slice profiles.
3. B2-noLLM adds a deterministic cross-layer supervisor that detects and resolves cross-tower conflicts without any LLM involvement.
4. B2 adds the guarded LLM strategic optimizer on top of the complete deterministic stack.

By measuring each layer independently, the contribution of the LLM layer can be disentangled from the deterministic architecture below it, preventing attribution errors that commonly inflate reported LLM contributions in network management literature.

1.6 Main Contributions

The main contributions of this project are:

- Design a hierarchical multi-agent system for joint handover and MEC control in 5G networks. The system coordinates fast reactive tower agents with a slow strategic optimizer using timescale separation and a shared intent vector.
- Propose a guarded LLM integration pattern in which the model selects from eight pre-validated policy bundles. A deterministic safety shield enforces bounded actuation, bundle rejection, rollback, and anti-oscillation constraints, ensuring that the LLM acts as a bounded advisor rather than an independent controller.
- Provide an experimental comparison of a telecom-domain fine-tuned LLM (TeleLLM, based on LLaMA-3-8B) against a general-purpose LLM backend (GPT-4o-mini) of comparable architectural scale. The comparison is designed to isolate the effect of domain specialization from model size and to evaluate both backends under identical prompt structure, bundle library, and safety constraints.

- An empirical analysis of the safety shield intervention rate, which modifies or reverses 40–62 percent of activated LLM selections in the canonical scenario, and a discussion of its implications for safe LLM deployment in telecommunications infrastructure.

1.7 Chapters layout

The remainder of this report is organized as follows:

Chapter One: Introduces the research background, problem statement, proposed solution, and main contributions.

Chapter Two: Reviews the theoretical background and related work relevant to 5G networks, MEC, LLM-based network management, and joint optimization approaches.

Chapter Three: Describes the simulation environment and the 3GPP-inspired digital twin design.

Chapter Four: Presents the proposed hierarchical multi-agent architecture and its operational components in detail.

Chapter Five: Covers the experimental methodology, evaluation metrics, results, and performance analysis.

Chapter Six: Concludes the report and discusses limitations and future work.

Chapter 2

Related Work

2.1 5G Network Architecture and Key Concepts

Modern 5G networks are built on a service-based architecture standardized by the 3rd Generation Partnership Project (3GPP). The radio access network is organized around gNBs, which communicate with UEs over the Uu interface and with each other over the Xn interface. The 5G core network is separated from the RAN, and the system supports network slicing, whereby logical end-to-end networks are instantiated over a shared physical infrastructure to serve different service requirements simultaneously [1].

Mobile Edge Computing (MEC) extends the compute and storage resources of the cloud to the edge of the network, physically co-located with the gNB. By executing computation tasks at the edge rather than routing them to a central cloud, MEC can dramatically reduce latency and backhaul traffic. However, MEC resources are limited, and their effective utilization requires intelligent admission control and load-balancing policies that account for the heterogeneous demands of coexisting service slices.

Handover is the process by which a UE transitions its radio connection from one gNB (the source) to another (the target) as it moves through the coverage area. In 5G, handover is predominantly triggered by A3 events, defined in 3GPP specifications as the condition in which a neighboring cell's Reference Signal Received Power (RSRP) exceeds that of the serving cell by a configurable hysteresis margin for a continuous time-to-trigger (TTT) duration. The hysteresis and TTT parameters govern the trade-off between handover aggressiveness (which improves radio quality but increases mobility signaling overhead) and handover conservatism (which reduces signaling but may allow UEs to remain on suboptimal cells).[8]

2.2 LLMs for Network Management and Intent-Based Networking

The adoption of LLMs in network management has grown rapidly since 2023. A comprehensive survey by Zhou et al. [9] finds that domain specialization generally assists performance, yet few works evaluate models on live network KPIs. Hong et al. [7] survey LLM applications in network design, automation, optimization, and security, observing that closed-loop control remains rare, with most deployments limited to static configuration or natural-language query answering.

On the intent-based networking side, Mekrache et al. [5] develop an LLM-based intent lifecycle management system for 5G/6G and demonstrate on a live EURECOM testbed that LLMs can decompose natural-language intents into network service descriptions. Dzeperoska et al. [10] consider LLM-assisted policy generation for intent-based management, identifying decomposition and lifecycle assurance as open problems. Jiang et al. [11] assemble an end-to-end intent-based networking framework that unites LLM-based intent translation with optimization. Each of these works demonstrates that LLM-based intent processing is feasible; none addresses how to safely close the optimization loop when LLM outputs are directly coupled to URLLC reliability and handover stability.

2.3 Multi-Agent LLM Frameworks with Validation

The Confucius framework [12], presented at ACM SIGCOMM 2025, implements intent-based network management at hyperscale using multi-agent LLMs with workflow decomposition, tool integration, retrieval-augmented memory, and regression-prevention validation. The present project shares the fundamental principle of structured multi-agent validation boundaries, applied to the specific problem of closed-loop joint handover and MEC optimization with a more targeted safety mechanism based on a bounded bundle library and a deterministic shield rather than a generic workflow engine.

NetLLM [13], presented at ACM SIGCOMM 2024, demonstrates that domain-specific fine-tuning of LLMs for networking is necessary for reliable network reasoning. NetConfEval [6] benchmarks LLM accuracy on network configuration tasks. Both works confirm that LLM outputs require structured evaluation against domain constraints, which is precisely the function performed by the safety shield in the proposed system.

2.4 Telecom-Domain Specialized Language Models

Continual pretraining of LLMs on telecom-specific corpora has emerged as a common method for improving performance on network-specific tasks. Maatouk et al. [19] introduce the Tele-LLMs family, models based on LLaMA-3 (1B to 8B parameters) further trained on Tele-Data, a curated corpus of 3GPP standards, IEEE papers, and telecom textbooks. The instruction-tuned LLaMA-3-8B-Tele-it model achieves improved performance on Tele-Eval benchmarks without sacrificing general-language capability, and this model serves as the TeleLLM backend in this project. Zou et al. [14] present TelecomGPT and demonstrate that domain-specific pretraining combined with direct preference optimization improves telecom math modeling and code generation. Kan et al. [15] introduce Mobile-LLaMA, showing that medium-sized fine-tuned models can compete with larger proprietary models in domain-specific tasks. The present project extends this literature to the closed-loop strategic control setting, where domain specialization must translate into improved operating regime selection.

2.5 Joint MEC Offloading and Handover Optimization

The joint optimization of MEC offloading and handover is a well-studied problem in 5G/6G research. Chen and Wang [16] apply decentralized deep reinforcement learning (DRL) to multi-user MEC offloading. Jahandar et al. [17] focus on mobility-aware offloading decisions in 5G with an emphasis on handover costs. Ho and Nguyen [18] jointly optimize server selection, cooperative MEC offloading, and handoff in a deep RL framework, demonstrating that coordinating all three decisions is essential. Zhou et al. [4] apply multi-agent DRL to joint multi-objective RAN slicing, demonstrating the inherent multi-metric complexity of slice optimization. Guo et al. [3] address joint handover control and power allocation using centralized training with decentralized execution MARL.

These reinforcement learning approaches perform well within their training distributions but encounter three structural limitations: they require substantial offline training in the target environment, they are brittle under distribution shift, and they produce policies that operators cannot easily inspect or explain. The LLM-based bundle selector proposed in this project does not require training in the target environment, is interpretable by

design (each bundle has a clear name and purpose), and is executed within a deterministic safety layer that bounds all network-facing parameter changes.

2.6 Positioning of This Work

Reviewing the complete literature, no existing work combines all four elements targeted in this project: joint handover and MEC optimization under multi-slice 3GPP traffic, a safety-guarded LLM at the strategic timescale, and a controlled comparison of a telecom fine-tuned model against a general-purpose backend of equal architectural scale, evaluated in reproducible form across multiple topologies and load conditions in a digital twin. Based on the reviewed literature, no existing work was found to combine all four of these elements within a single system and evaluation framework.

Chapter 3

Simulation Environment

3.1 Digital Twin Design

The simulation environment is a stochastic, reproducible digital twin modeled after 3GPP standards. It is designed to maintain physical interpretability of the radio system and computing mechanism while providing sufficient control over the agentic system to support rigorous ablation studies. The digital twin is not a full 3GPP protocol simulator; it is a random-plus-deterministic network simulation with explicit topology, random workload generation via explicit seeding, and explicit outcome tracking for task offloading and handover events. All randomness in the simulation is seeded, which ensures that different control modes can be compared fairly under identical traffic and mobility conditions.

3.2 Radio and Mobility Model

The simulator implements the 3GPP TR 38.901 [8] urban macro (UMa) path-loss model at a carrier frequency of 3.5 GHz with 100 MHz bandwidth per gNB and 46 dBm maximum transmit power. The radio model computes received signal strength, interference, and effective SINR for every UE at every 1 ms simulation step to support channel quality estimation, handover control, and MEC scheduling decisions.

The selected 3.5 GHz carrier frequency represents a commonly deployed mid-band 5G spectrum range that provides a balance between coverage and capacity, while the 100 MHz bandwidth reflects a typical enhanced Mobile Broadband (eMBB) deployment configuration in 5G NR systems.[1] [8]

The three-dimensional UE-to-gNB distance is computed from the ground-plane distance d_{2D} and antenna heights: [8]

$$d_{3D} = \sqrt{d_{2D}^2 + (h_{BS} - h_{UT})^2} \quad (3.1)$$

Where d_{2D} is the ground-plane UE-gNB distance in meters, $h_{BS} = 25$ m is the base station antenna height, and $h_{UT} = 1.5$ m is the UE antenna height.

The breakpoint distance, which determines the transition between LOS and NLOS path loss regimes, is: [8]

$$d_{BP} = 4 \cdot h_{BS} \cdot h_{UT} \cdot \frac{f_c}{c} \quad (3.2)$$

Where $f_c = 3.5$ GHz and $c = 3 \times 10^8$ m/s.

LOS and NLOS path loss are computed as: [8]

$$PL_{LOS} = 28.0 + 22 \cdot \log_{10}(d_{3D}) + 20 \cdot \log_{10}(f_c) \text{ [dB]} \quad (3.3)$$

$$PL_{NLOS} = \max(PL_{LOS}, 13.54 + 39.08 \cdot \log_{10}(d_{3D}) + 20 \cdot \log_{10}(f_c) - 0.6(h_{UT} - 1.5)) \text{ [dB]} \quad (3.4)$$

Equation (3.3) applies when $d_{3D} < d_{BP}$ (LOS regime);

Equation (3.4) applies when $d_{3D} \geq d_{BP}$ (NLOS regime).

The Reference Signal Received Power (RSRP), noise floor, and SINR are: [8]

$$RSRP = P_{TX} - PL + \xi \text{ [dBm]} \quad (3.5)$$

$$N = N_0 + 10 \cdot \log_{10}(B) + NF \text{ [dBm]} \quad (3.6)$$

$$SINR = S / (I + N) \quad (3.7)$$

Where $P_{TX} = 46$ dBm, PL is from Equations (3.3)– (3.4), ξ is spatially correlated log-normal shadow fading (Equation 3.8), $N_0 = -174$ dBm/Hz, $B = 100$ MHz, $NF = 5$ dB, S is serving-cell received power, and I is aggregate inter-cell interference.

Shadow fading is modeled with $\sigma_{sf} = 8$ dB standard deviation and $d_{corr} = 50$ m decorrelation distance to capture large-scale signal fluctuations caused by obstacles and environmental variations.

The per-step fading update and spatial correlation coefficient are: [8]

$$\xi_n = \rho \cdot \xi_{n-1} + \sqrt{1 - \rho^2} \cdot \sigma_{sf} \cdot w \quad (3.8)$$

$$\rho = \exp\left(-\frac{\Delta d}{d_{corr}}\right) \quad (3.9)$$

here Δd is the UE displacement since the last step and $w \sim N(0,1)$.

Effective channel throughput is estimated with a Shannon approximation and an implementation efficiency factor $\eta = 0.75$:

$$C = \eta \cdot B \cdot \log_2(1 + SINR) \quad [Mbps] \quad (3.10)$$

The efficiency factor η accounts for pilot overhead, guard subcarriers, and HARQ retransmission losses.

3.3 Network Topology

Two topology configurations are used. The 3-gNB layout places gNBs in a triangular arrangement with 500 m inter-site distance (ISD). The 5-gNB layout uses a pentagonal arrangement with 400 m ISD, expanding the coverage are. The two topologies are intended to evaluate the system under different network density and scalability conditions.

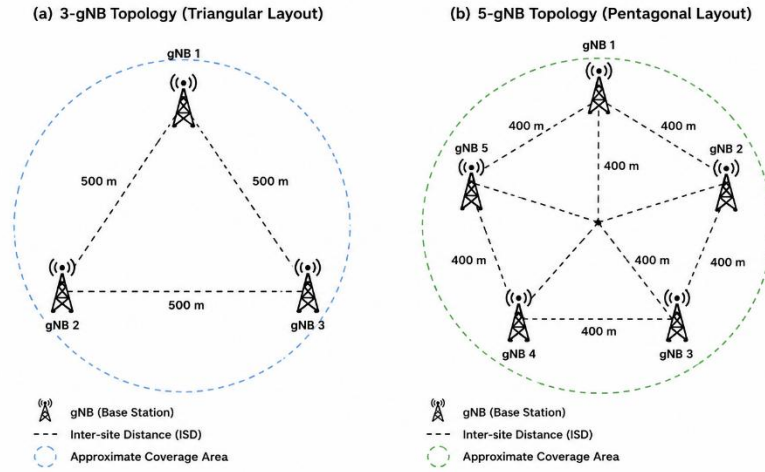


Fig. 3.1. Network topology configurations used in the simulation

UE initial positions are randomly assigned within the coverage area, and UE mobility follows a Gauss-Markov model for the full 600-second simulation duration.

Inter-gNB task migration uses the Xn interface with a latency model that accounts for both fixed switching overhead and distance-dependent propagation:

$$t_{lat} = 1.0 + \left(\frac{d_{Xn}}{1000} \right) \cdot 0.4 \text{ [ms]} \quad (3.11)$$

where d_{Xn} is the inter-gNB Euclidean distance in meters, 1.0 ms is fixed switching overhead, and 0.4 ms/km represents distance-dependent transport delay over the Xn/backhaul path.

3.4 MEC Compute Model

Each gNB hosts a local edge compute unit. MEC capacity varies by topology: 100 GFLOPS for the 3-gNB configuration, and 150–200 GFLOPS for the 5-gNB configuration. Task execution follows a priority-ordered placement policy: local edge first; Xn peer when the deadline allows and load permits; regional MEC (15 ms latency, 1 Gbps backhaul) for non-URLLC tasks; and task drop otherwise. The total Xn migration cost is:

$$t_{mig} = t_{lat} + \frac{P_{MB} \cdot 8}{BW_{Xn} \cdot 1000} + t_q \text{ [ms]} \quad (3.12)$$

where t_{lat} is from Equation (11), P_{MB} is the task payload in MB, $BW_{Xn} = 10$ Gbps is the Xn backhaul bandwidth, and $t_q = 0.5$ ms is the queue-entry delay.

Device energy is tracked through a composite proxy metric that captures three contributing components: local compute energy, peer compute energy with Xn overhead, and the energy penalty associated with unnecessary handovers:

$$E_{proxy} = G_{local} \cdot 1.0 + G_{peer} \cdot 1.3 + n_{UHO} \cdot 75 \text{ [mJ]} \quad (3.13)$$

where G_{local} and G_{peer} are completed GFLOPS executed locally and on an Xn peer respectively, 1.0 mJ/GFLOP is the local compute energy coefficient, 1.3 mJ/GFLOP is the Xn-peer coefficient (reflecting 30% overhead for radio re-establishment and forwarding), n_{UHO} is the unnecessary handover count, and 75 mJ/HO is the per-handover energy penalty derived from implementation measurements. Equation (13) is a proxy evaluation metric for comparative analysis and is not a first-principles battery model.

URLLC tasks have an 8 ms execution deadline and a compute demand of 1–5 GFLOPS, arriving with a 15% per-millisecond probability. eMBB tasks arrive as a Poisson process dependent on traffic urgency state. mMTC tasks arrive in periodic bursts characteristic of sensor reporting patterns.

3.5 Multi-Slice Traffic Model

The UE population is divided into three service families: URLLC devices (vehicle OBU, industrial controller), eMBB devices (smartphone, tablet, XR headset, laptop), and mMTC devices (wearable, IoT sensor). The device class distribution across scenarios is load-dependent: in the standard 15-UE canonical scenario, the class distribution is 20% URLLC, 50% eMBB, and 30% mMTC; in the 45-UE scalability scenario, the population is tripled proportionally.

The intent engine generates a repeatable sequence of canonical events for each scenario seed, including planned mobility transitions, planned congestion bursts, and planned slice-priority changes, coupled with random intent activation per seed. This hybrid generation technique ensures that each seed produces a distinct traffic realization while retaining a common structural template that makes cross-seed comparisons meaningful.

3.6 Time Model and Telemetry Cadence

The proposed simulation uses a multi-tier time model with four distinct timescales: physics and radio computation at 1 ms resolution; HARQ and queue adjustments at 10 ms; telemetry aggregation and agent decision-making at 100 ms; and strategic LLM optimization at 60-second intervals (with deterministic trigger evaluation at each scheduled call). The 100 ms global telemetry snapshots comprise per-UE state vectors, per-gNB utilization, slice-level KPIs, and task/handover outcome records, and constitute the primary input to the agent federation. Fig. 3.2 illustrates the multi-timescale hierarchical control architecture.

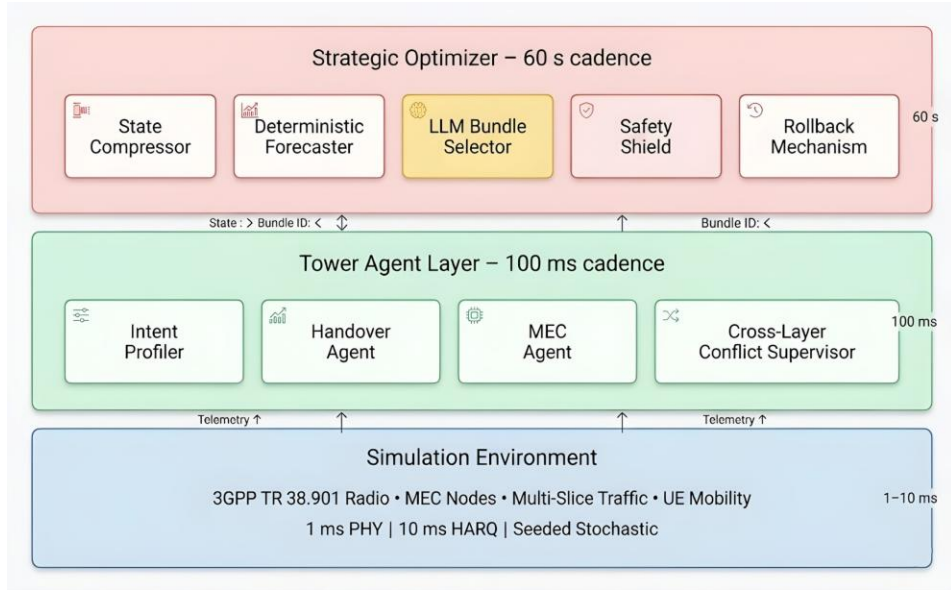


Fig. 3.2. Multi-Timescale Hierarchical Control Architecture

This hierarchical timing arrangement remains fixed throughout all simulation runs. The same telemetry cadence and control update structure are preserved across all evaluated modes to maintain consistency in the observation and decision process.

Chapter 4

Hierarchical Agent Architecture

4.1 Design Principles

The architectural design is governed by two core principles. The first is Temporal Separation of Responsibility: safety-critical behaviors such as admission control, handover execution, MEC task placement, and conflict resolution operate within the 1–100 ms timescale using deterministic rules and bounded heuristics. The LLM only enters at the 60-second strategic timescale, where its inference latency has no impact on radio-layer safety, and its output is validated before any parameter change reaches the network. The second principle is Intent-Mediated Coordination: rather than each component independently accessing raw telemetry and risking conflicting actions, all components consume a shared structured intent vector. This ensures that any strategic alteration propagates coherently through both handover and MEC reasoning without requiring direct communication links between components. Fig. 4.1 illustrates the complete end-to-end hierarchical control flow.

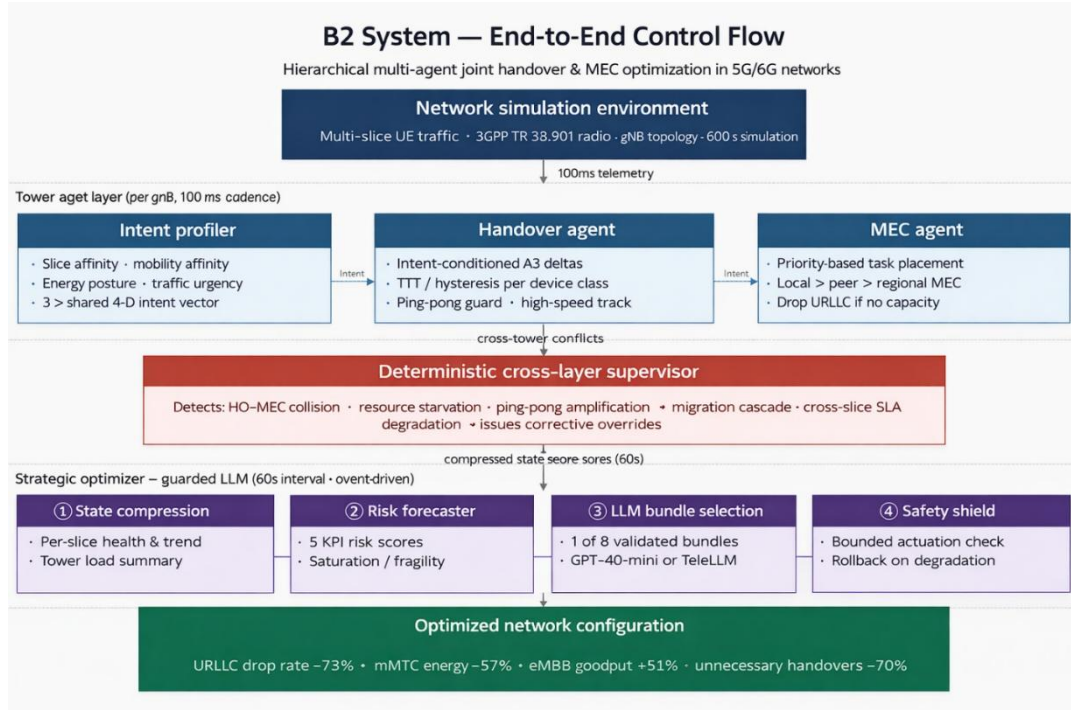


Fig. 4.1. End-to-End Hierarchical Control Flow of the Proposed System

4.2 Control Mode Hierarchy

The framework defines four modes of operation, each a strict superset of the previous. Table 4.1 summarizes the components and operational role of each mode.

Table 4.1: Control Mode Hierarchy

Mode	Components	Role
B0 (Baseline)	A3 handover, greedy MEC	Uncontrolled reference: standard 3GPP handover with default greedy MEC scheduling.
B1 (Intent-Aware)	B0 + tower agents + intent profiler	Per-gNB tower agents process 100 ms telemetry through intent-conditioned handover and MEC logic.
B2-noLLM	B1 + deterministic supervisor	Cross-layer conflict detector resolves six conflict archetypes heuristically.
B2 (Guarded LLM)	B2-noLLM + strategic optimizer	LLM selects from eight named policy bundles under a deterministic safety shield with rollback support.

This staging serves not only operational deployment convenience but also the evaluation methodology. By measuring the contribution of each layer independently, the value added by the LLM can be rigorously disentangled from the deterministic architecture on which it is built.

4.3 Intent Profiler and Vector Representation

Each UE's state is mapped to a structured intent vector $v = (s_U, s_E, s_M, m_{stat}, m_{ped}, m_{veh}, m_{hi}, u, \varepsilon) \in [0,1]^9$ that compactly encodes slice class, mobility state, traffic urgency, and energy posture into a single structured representation consumed by both the handover and MEC agents.

The slice affinity s_k over {URLLC, eMBB, mMTC} aggregates eight observable signals with fixed weights, normalized to a probability simplex:

$$s_k = \sum w_i \cdot \varphi_{\{i,k\}}, \quad \sum s_k = 1 \quad (4.1)$$

where w_i are fixed weights summing to 1 and $\varphi_{\{i,k\}} \in [0,1]$ are observable signal values. Mobility affinity places the UE's current speed on a soft-membership scale over {stationary, pedestrian, vehicular, high-speed} using Gaussian kernels centered at speed prototypes $c_k \in \{0.0, 1.5, 7.0, 20.0\}$ m/s:

$$\mu_k(v) = \exp\left(-\frac{(v - c_k)^2}{2\sigma^2}\right) \quad (4.2)$$

where v is the current UE speed in m/s and $\sigma = 5.0$ m/s.

Traffic urgency combines normalized queue depth, drop rate, and an activity-hint offset:

$$u = \text{clip}(0.6 \cdot q + 0.4 \cdot \tilde{d} + \delta_{act}, [0,1]) \quad (4.3)$$

where $q \in [0,1]$ is normalized queue depth, $\tilde{d} = \text{clip}\left(\frac{n_{drop}}{10}\right)$,

and $\delta_{act} \in \{-0.25, -0.10, +0.20, +0.30\}$ encodes the activity class (idle, background, streaming, interactive).

Energy posture reflects the UE's power budget through a five-signal weighted combination:

$$\varepsilon = \text{clip}(\sum w_j \cdot e_j, [0,1]) \quad (4.4)$$

where signals e_j and weights w_j cover device prior (0.65), battery class (0.125), compute mean (0.10), packet IAT (0.075), and activity hint (0.05). A value of $\varepsilon = 1$ indicates a performance-hungry device; $\varepsilon = 0$ indicates an energy-conservative device.

All four intent dimensions are smoothed across the 100 ms telemetry window using an Exponential Moving Average:

$$\hat{v}_t = \alpha \cdot v_t + (1 - \alpha) \cdot \hat{v}_{\{t-1\}}, \quad \alpha = 0.3 \quad (4.5)$$

where $\alpha = 0.3$ gives an effective window of approximately 10 snapshots (1 second). The handover and MEC agents read this smoothed intent vector as their sole input at every 100 ms control window.

4.4 Tower-Level Agents

Each tower-level agent performs profiler update, handover management, MEC scheduling, supervisor check, and optimizer policy application every 100 ms. The handover agent operates on the 3GPP A3 event: a handover is triggered when a neighbor cell's RSRP

exceeds the serving cell's RSRP by the hysteresis margin and the condition holds continuously for the full time-to-trigger duration:

$$RSRP_{nbr(t')} > RSRP_{srv(t')} + hys, \quad \forall t' \in [t - TTT, t] \quad (4.6)$$

where the quantifier $\forall t' \in [t - TTT, t]$ requires the signal condition to hold at every point in the TTT window, not just at time t . The hysteresis and TTT parameters are intent-conditioned per Equations (20) and (21): URLLC UEs receive aggressive settings (low hys, low TTT) to remain on the strongest cell; mMTC UEs receive conservative settings to minimize unnecessary migrations; high-speed UEs receive short TTT to track mobility. Persistent ping-pong history adds a protective extra hysteresis. The continuous affine parameter maps, clamped to safe ranges, are:

$$hys = 3.0 - 0.5 \cdot s_U + 2.0 \cdot s_M - 0.5 \cdot u + 2.0 \cdot 1[n_{pp} \geq 2] + \Delta hys_{LLM} [dB, 1 - 6] \quad (4.7)$$

$$TTT = 160 - 40 \cdot s_U + 160 \cdot s_M - 80 \cdot m_H + 80 \cdot m_S + \Delta TTT_{LLM} [ms, 40 - 320] \quad (4.8)$$

where $\Delta hys_{LLM} \in [-2, +2]$ dB and $\Delta TTT_{LLM} \in [-40, +40]$ ms are the parameter deltas applied by the active LLM bundle. The MEC agent assigns each task a scheduling priority score:

$$P(\tau) = 50 + 30 \cdot s_U + 10 \cdot s_E + 15 \cdot u + 5 \cdot \varepsilon + 20 \cdot \max\left(0, 1 - \frac{dl_\tau - t}{500}\right) \in [50, 100] \quad (4.9)$$

where dl_τ is the task deadline in ms, t is the current tick in ms, and the final term gives additional urgency in the last 500 ms before deadline. High-urgency classes raise effective Xn-peer headroom; energy-conservative devices prefer local execution unless deadline pressure forces migration.

4.5 Deterministic Cross-Layer Supervisor

The B2-noLLM supervisor evaluates six types of cross-layer conflicts at every 100 ms control window: (a) handover/MEC collision, where a pending handover would disrupt an active MEC task chain; (b) resource starvation, where a posture favoring one slice disadvantages another; (c) migration cascade, where successive task migrations increase cumulative latency across the mesh; (d) ping-pong amplification, where handover oscillation destabilizes mMTC energy; (e) energy/latency trade-off conflicts, where posture choices benefiting one slice harm another; and (f) cross-slice SLA degradation.

Upon detecting a conflict, the supervisor issues directive policy overrides to the affected tower agents — for example, temporarily freezing handovers when they would interrupt URLLC tasks, or throttling eMBB admission to prevent mMTC starvation. This supervisor is critical because it resolves cross-tower conflicts that tower agents cannot detect in isolation, as each tower agent has no visibility into the state or planned actions of neighboring towers.

4.6 Strategic B2 Optimizer

The strategic optimizer activates every 60 seconds and proceeds through four phases.

In the state compression phase, the last system state is compressed into a structured summary including global slice health, trend characteristics (improving, degrading, or stable), cross-tower pressure imbalance, and a pre-filtered bundle shortlist generated by deterministic scoring.

In the forecasting phase, a lightweight deterministic forecaster projects five risk scores for the coming window: MEC saturation risk, load change pressure, URLLC fragility, mMTC energy risk, and mobility pressure. Each score is normalized to $[0,1]$ and labeled with a qualitative designation (low, medium, high).

In the LLM bundle selection phase, the state summary and forecast are sent to the LLM with a structured prompt. The LLM selects one bundle from the eight-element library. The optimizer also maintains an experience buffer of up to three recent state-action-outcome pairs from the current run, which are included in the prompt as few-shot examples.

In the safety validation phase, the shield validates the LLM's selection against hard invariants. If validation fails, the system falls back to the last known safe bundle without LLM involvement.

The LLM is activated only when the state summary indicates that a strategic change has value: increasing MEC saturation, high mobility pressure, significant load imbalance across towers, or recurring instability. If none of these conditions apply, the current bundle remains active and the LLM call is skipped. This conditional triggering reduces unnecessary LLM invocations, which is important in deployments where inference has a cost.

The available strategic policy bundles and their corresponding network actions are summarized in Table 4.2.

Table 4.2: LLM Policy Bundle Library

Bundle	Actions Applied
BALANCED	Applies mild TTT tightening and a low eMBB scheduling boost; enables conservative Xn coordination.
PROTECT_URLLC	Freezes handover parameters; blocks MEC migration; enforces IoT sensor hard floor at maximum strictness.
FAVOR_EMBB	Aggressively tightens TTT for eMBB UEs; raises their MEC scheduling priority; enables Xn compute migration.
SAVE_MMTc_ENERGY	Widens TTT/hys to suppress unnecessary handovers; throttles mMTC compute access to reduce radio reconnect cycles.
MOBILITY_STABILIZE	Strongly widens TTT and hys to stop handover churn; forces local compute; disables cross-tower rebalance.
CONGESTION_RELIEF	Tightens TTT to offload eMBB UEs faster; redistributes MEC load across Xn peers; raises eMBB compute priority.
PREEMPTIVE_REBALANCE	Proactively migrates eMBB tasks to the least-loaded Xn peer; applies maximum cross-tower rebalance before saturation.
RECOVERY_MODE	Restores the last safe policy snapshot; loosens handover thresholds; blocks all migration to limit further disruption.

Each bundle maps to bounded parameter adjustments validated against physical and service constraints. The bundle selection boundary eliminates two problems that frequently destabilize learning-based network control: action space explosion (continuous parameter control can drift into unsafe regions) and non-reproducible

oscillations (LLM sampling stochasticity becomes bounded rather than arbitrary when the output is a discrete bundle ID passed through deterministic validation).

4.7 Safety Shield and Rollback

The safety shield enforces three categories of constraints. Bounded actuation ensures that all bundle-induced parameter changes remain within predefined safe ranges: TTT deltas within ± 40 ms of the A3 baseline, hysteresis deltas within ± 2.0 dB. Bundle rejection rejects any selection that would violate hard constraints, for example, selecting FAVOR_EMBB when URLLC fragility exceeds a critical threshold, falling back to BALANCED. Rollback and anti-oscillation trigger a reversion to the previous stable bundle when monitored KPIs degrade beyond a threshold in two consecutive 100 ms windows. The formal rollback condition is:

$$rollback(b) \leftarrow 1[\Delta KPI(t) > \rho_s] + 1[\Delta KPI(t-1) > \rho_s] = 2 \quad (4.10)$$

where $\Delta KPI(t) = (KPI(t) - KPI_{ref}) / KPI_{ref}$ is the relative degradation, $\rho_s \in [0,1]$ is a per-bundle rollback sensitivity parameter, and $1[\cdot]$ is the indicator function. The three monitored KPIs are: URLLC drop rate (rollback if delta exceeds $\max(0.15 \text{ pp}, 0.35 \cdot \rho_s)$); IoT sensor task completion (rollback if below 82% hard floor or drops more than $\max(1.0 \text{ pp}, 1.4 \cdot \rho_s)$); eMBB goodput is excluded from the rollback signal due to high short-term variance. After executing a bundle, the shield holds it for 1–2 strategic cycles (60 s each) before accepting a switch, preventing rapid oscillation driven by sampling stochasticity.

Chapter 5

Results and Analysis

5.1 Benchmark Scenarios

Four evaluation scenarios are defined, covering various combinations of network topology and traffic load. Each scenario runs for 600 seconds of simulated time with seeded randomness:

Table 5.1: Benchmark Scenario Configuration

Scenario	Topology	UEs	Seeds	Runs/Mode
Canonical	3 gNB triangular macro, 500 m ISD	15	8	40
Scalability A	3 gNB triangular macro, 500 m ISD	45	4	20
Scalability B	5 gNB pentagonal macro, 400 m ISD	30	4	20
Scalability C	5 gNB pentagonal macro, 400 m ISD	45	4	20

The canonical scenario uses eight seeds to achieve greater statistical power for the primary comparisons. Each scalability scenario uses four seeds. The five-mode evaluation matrix (B0, B1, B2-noLLM, B2-GPT-4o-mini, B2-TeleLLM) results in a total of 100 independent simulation runs across all scenarios and modes.

5.2 Evaluation Metrics

Seven metrics are reported across the three service families:

For URLLC: P99 end-to-end latency (ms, lower is better) and task drop rate (percentage of tasks missing the 8 ms deadline, lower is better). The drop rate is the primary URLLC safety metric.

For eMBB: task completion rate (percentage, higher is better) and cumulative goodput (GFLOPS/s, higher is better). Goodput is defined as the total completed eMBB compute in GFLOPS across all eMBB device classes divided by the simulation duration:

$$GP_{eMBB} = \Sigma \frac{G_k}{T_{sim}} \text{ [GFLOPS/s]} \quad (5.1)$$

where G_k is completed GFLOPS for eMBB device class k and $T_{sim} = 600$ s. This measures real computation throughput delivered to eMBB users, not radio-layer bit rate.

For mMTC: task completion rate (percentage, higher is better, expected near 100%), device energy consumption using the E_{proxy} metric in mJ (lower is better), and unnecessary handover count (lower is better). Two mMTC device subtypes are reported independently: IoT sensor completion percentage and wearable completion percentage.

5.3 LLM Backends

The B2 mode compares two LLM backends. The first is GPT-4o-mini, accessed via the OpenAI API, a proprietary general-purpose model that has been independently benchmarked as comparable to open-source models in the LLaMA-3-8B parameter class. The second is TeleLLM (AliMaatouk/LLaMA-3-8B-Tele-it [19]), an instruction-tuned LLaMA-3-8B model further trained on a curated telecom corpus comprising 3GPP standards, IEEE publications, and telecom textbooks, run on local GPU infrastructure. Both backends use identical prompt structure, state summarization pipeline, bundle library, and safety shield. The model weights are the only difference. Both backends are called with temperature 0.7 and no top-p filtering.

The comparison is designed to be controlled for model scale, allowing domain-specific pretraining to serve as the primary explanatory variable for any observed performance difference. However, it is acknowledged that the two backends also differ in their instructional tuning strategy and inference infrastructure, so domain specialization

cannot be claimed as the sole confounding variable. LLM inference latency is not included in the reported KPIs because the two backends operate on different infrastructure (cloud API versus local GPU) and what matters for the evaluation is whether each backend selects good bundles, not how quickly it selects them.

5.4 Statistical Reporting

Results are reported as mean values across seeds for each scenario-mode combination. Standard deviations are included where variability is relevant to interpreting the results. Relative differences between reference modes are expressed using delta notation (Δ). Scenario-level leadership counts, indicating how many of the four scenarios each mode achieves the best value on each metric, provide a cherry-picking-resistant cross-scenario summary. A paired t-test across the eight canonical seeds is used to assess statistical significance for the primary URLLC drop comparison between B2-TeleLLM and B2-noLLM.

5.5 Deterministic Layer Progression

The primary finding of this study is not which LLM backend is superior. It concerns the clarity with which each control layer adds measurable value on top of the layer below. Understanding this progression is essential before interpreting the LLM comparison, because it establishes that the deterministic layers are the dominant source of improvement. Fig. 5.1 shows the cross-scenario KPI comparison across all control modes.

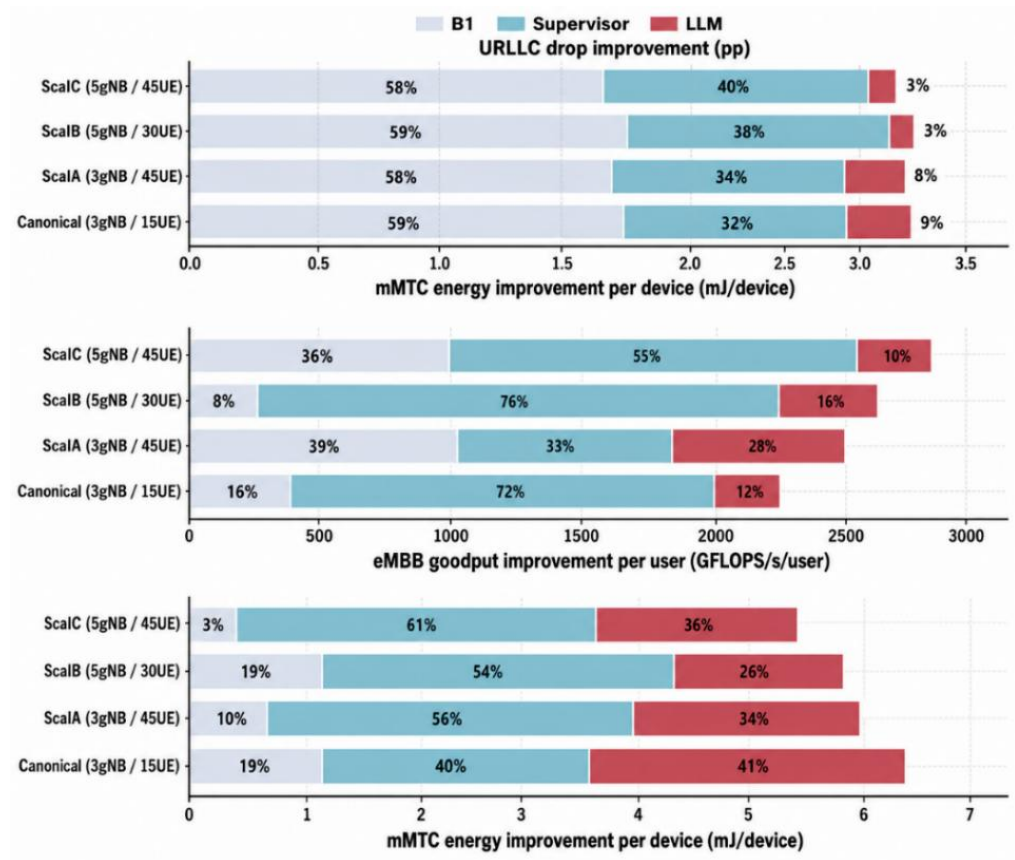


Fig. 5.1. Cross-Scenario KPI Comparison Across Control Modes

Table 5.2: Deterministic Layer Progression — B0→B1→B2-noLLM Deltas

Scenario	B1→B0 URLLC Δ (pp)	B1→B0 Goodput Δ (GFLOPS/s)	B1→B0 Energy Δ (mJ)	B2n→B1 URLLC Δ (pp)	B2n→B1 Goodput Δ (GFLOPS/s)	B2n→B1 Energy Δ (mJ)
3 gNB / 15 UE	-1.890	+8.69	-1,201	-1.019	+17.82	-5,381
3 gNB / 45 UE	-1.846	+7.93	-14,513	-1.094	+44.63	-12,375
5 gNB / 30 UE	-1.940	+13.56	-2,087	-1.266	+38.13	-19,538
5 gNB / 45 UE	-1.780	+2.84	-17,448	-1.222	+50.95	-26,433

B1 versus B0: Intent-aware tower agents reduce URLLC drop by 1.78–1.94 pp in all four scenarios. This occurs because the agents adapt TTT and hysteresis parameters according to each UE's slice affinity: URLLC devices receive aggressive handover parameters to remain on the strongest cell, while mMTC devices receive conservative settings to minimize unnecessary mobility. The eMBB goodput gain at B1 is modest (2.8–13.6 GFLOPS/s) because intent-awareness at this layer primarily stabilizes the radio layer rather than optimizing computation.

B2-noLLM versus B1: The deterministic supervisor introduces a second independent improvement. The most significant impact is on eMBB goodput, with improvements of 17.8, 44.6, 38.1, and 50.9 GFLOPS/s in the four scenarios respectively. This large improvement arises from resolving handover/MEC collision events, where a pending handover interrupt would disrupt an active MEC task chain. Such collisions cannot be prevented by B1 alone, since each tower agent operates independently without knowledge of neighboring towers. The energy savings relative to B1 (5,381–26,433 mJ) result from successful suppression of mMTC ping-pong events.

A cross-layer trade-off emerges at scale: in the heavy-load 3 gNB / 45 UE scenario, B2-noLLM reduces URLLC drop significantly but also increases unnecessary handovers from 351 (B1) to 387. This occurs because the supervisor's aggressive congestion relief redistributes load by inducing handovers that benefit compute distribution but damage mMTC mobility stability. This is corrected in the LLM modes: TeleLLM reduces unnecessary handovers to 251, GPT-4o-mini to 280, demonstrating that strategic bundle selection can resolve trade-offs that the deterministic supervisor introduces independently.

5.6 LLM-Assisted Optimization versus Deterministic Baseline

Table 5.3 shows the per-scenario deltas for both LLM backends relative to B2-noLLM, representing the incremental contribution of the guarded LLM selection layer.

Table 5.3: B2 Backend Deltas vs. B2-noLLM

Scenario	Backend	Δ URLLC drop (pp)	Δ eMBB comp (pp)	Δ Goodput (GFLOPS/s)	Δ Energy (mJ)	Δ mMTC comp (pp)
3/15 (canonical)	GPT-4o-mini	-0.253	+1.64	+17.89	-689	-0.49
3/15 (canonical)	TeleLLM	-0.285	+1.99	+18.24	-866	+0.77
3/45	GPT-4o-mini	-0.053	+1.64	+26.79	-7,877	+1.75
3/45	TeleLLM	-0.246	+2.11	+27.29	-10,389	+1.75
5/30	GPT-4o-mini	-0.018	+1.71	+10.77	-2,795	+1.00
5/30	TeleLLM	-0.085	+1.71	+18.47	-4,175	+1.17
5/45	GPT-4o-mini	-0.018	+1.51	+29.35	-3,743	+2.69
5/45	TeleLLM	-0.087	+1.82	+29.85	-4,607	+2.69

Both backends improve eMBB completion, goodput, and mMTC energy in every scenario. The key finding is that, regardless of topology or load, guarded LLM bundle selection consistently adds value on top of the deterministic baseline. TeleLLM achieves larger URLLC drop reductions and larger energy savings in all four scenarios, while GPT-4o-mini

matches or marginally leads on eMBB metrics. The scalability scenarios use four seeds each, so directional consistency is reliable but precise magnitudes of inter-backend differences in those scenarios should be interpreted with caution.

5.7 Cross-Scenario Metric Leadership

To prevent cherry-picking of favorable scenarios, scenario-level metric leadership counts are reported: the number of scenarios (out of four) in which each mode achieves the best value on each metric.

Table 5.4: Scenario-Level Metric Leadership Count (Out of 4 Scenarios)

Mode	URLLC drop	URLLC P99	eMBB comp	eMBB goodput	mMTC comp	IoT sensor	Wearable	Energy	Unnecessary HO	Total
B0	0	0	0	0	0	0	0	0	0	0
B1	0	0	0	0	0	2	1	0	0	3
B2-noLLM	0	0	0	0	0	0	0	0	0	0
B2-GPT-4o-mini	0	1	2	4	1	0	3	0	1	12
B2-TeleLLM	4	3	2	0	3	2	0	4	3	21

This table provides the most candid view of the comparative results. B0 and B2-noLLM do not dominate any metric in any scenario. B1 leads in three scenarios, exclusively on device-class subtypes (IoT sensor and wearable), where its locality-preserving default behavior is incidentally effective. The meaningful separation is between the two LLM backends: TeleLLM provides better URLLC reliability, mMTC energy, and mMTC

completion, which are the more challenging objectives under mobility and contention. GPT-4o-mini leads on eMBB goodput and wearable completion. The pattern is consistent: domain fine-tuning benefits reliability and energy metrics, while general-purpose training benefits throughput metrics.

Fig. 5.2 plots the joint URLLC drop and mMTC energy operating point for every mode-scenario combination, making the Pareto improvement of each successive control layer visible.

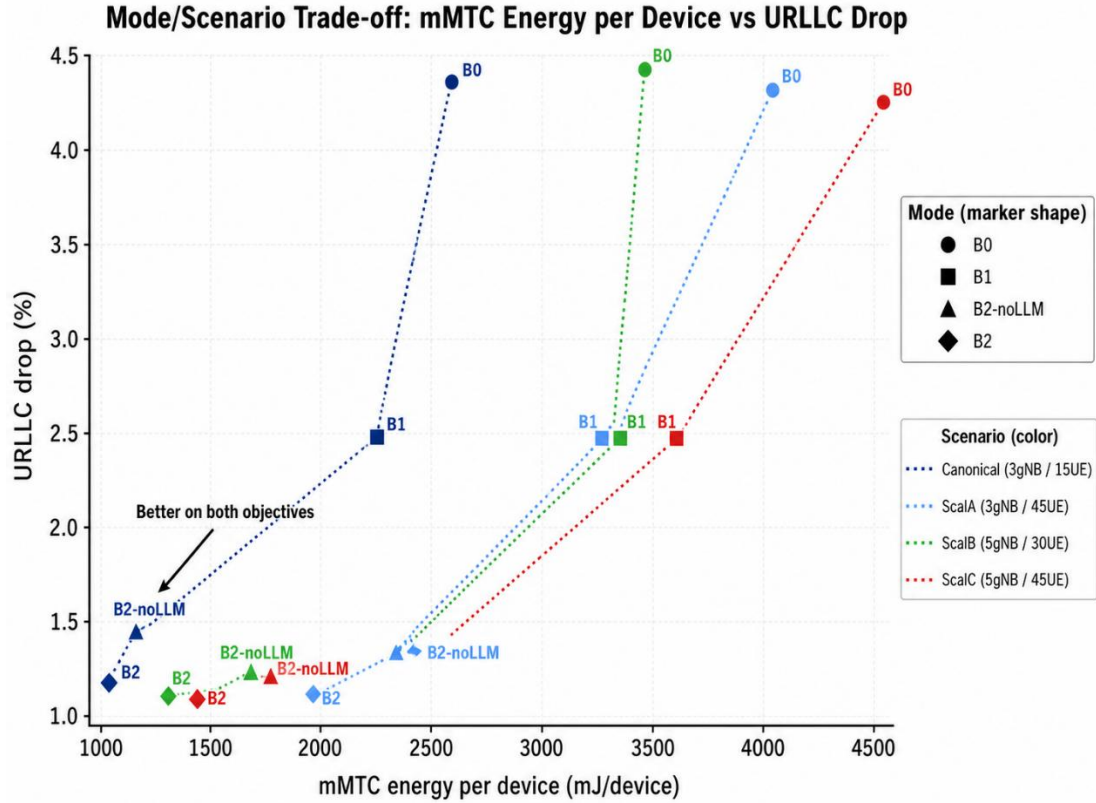


Fig. 5.2. URLLC Drop vs. mMTC Energy Trade-Off Across Modes and Scenarios

5.8 Canonical Scenario: Detailed Analysis

The canonical 3 gNB / 15 UE scenario, with its eight seeds, provides the greatest statistical power for detailed comparison. Table 5.5 presents the full KPI summary for this scenario.

Table 5.5: Canonical 3 gNB / 15 UE — Full KPI Summary (8-Seed Average)

Mode	URLLC drop %	URLLC P99 ms	eMBCOMP %	eMBB goodput GFLOPS/s	mMTC energy mJ	mMTC comp %	IoT sensor %	Wearable %	Unnecessary HO
B0	4.365	4.596	84.79	404.87	13,143	89.20	86.79	91.59	57.65
B1	2.476	4.593	87.23	413.57	11,942	89.48	87.09	91.84	30.82
B2-noLLM	1.457	4.582	87.33	431.39	6,561	89.50	87.20	91.77	26.96
B2-GPT-4o-mini	1.204	4.572	88.97	449.27	5,872	89.01	87.05	90.90	20.90
B2-TeleLLM	1.172	4.558	89.32	449.62	5,695	90.27	90.09	90.38	17.17

Starting from B0, URLLC drop falls from 4.365% to 1.172% with B2-TeleLLM, a 73% relative reduction across all three control layers. Decomposing this: B1 contributes 1.89 pp (59% of the total 3.19 pp reduction); B2-noLLM adds 1.02 pp (32%); and the LLM layer contributes 0.285 pp (9%) on top of the deterministic foundation. The LLM's contribution, while smaller, is statistically significant in the canonical scenario and consistent in direction across all four scenarios. Unnecessary handovers decrease from 57.65 to 17.17, a 70% reduction. mMTC energy decreases from 13,143 to 5,695 mJ, a 57% reduction.

Regarding statistical significance: the standard deviation of B2-TeleLLM URLLC drop across eight seeds is 0.023 pp, while that of B2-noLLM is 0.019 pp. The 0.285 pp difference

is substantially larger than the noise floor. A paired t-test across the eight canonical seeds yields $p < 0.01$ for B2-TeleLLM versus B2-noLLM on URLLC drop, confirming statistical significance. Goodput varies more across seeds (standard deviation of 3.2 GFLOPS/s with B2-TeleLLM), but the 18.24 GFLOPS/s improvement over B2-noLLM represents approximately 5.7 standard deviations and is clearly significant. The mMTC energy standard deviation of 142 mJ makes the 866 mJ gain (6.1 standard deviations) equally conclusive.

URLLC P99 latency changes only marginally across modes, from 4.596 ms (B0) to 4.558 ms (B2-TeleLLM), less than 1% compression. This is informative: the URLLC improvement comes from reducing the number of tasks that miss their 8 ms deadline, not from compressing the underlying latency distribution. The fixed task deadline structure determines the latency distribution; the control layers work to avoid handover interruptions and MEC queueing events that cause deadline violations, rather than to modify the latency envelope itself.

Fig. 5.3 and fig. 5.4 show the URLLC P99 latency and mMTC task completion metrics across all scenarios. Fig. 5.5, fig. 5.6, and fig. 5.7 disaggregate results to device-class level.

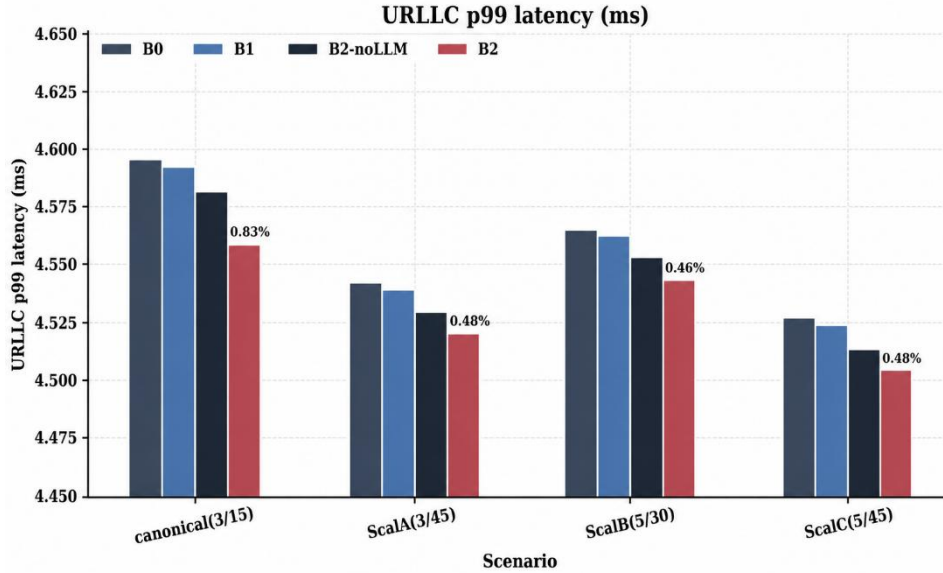


Fig. 5.3. URLLC P99 Latency Across Scenarios and Control Modes

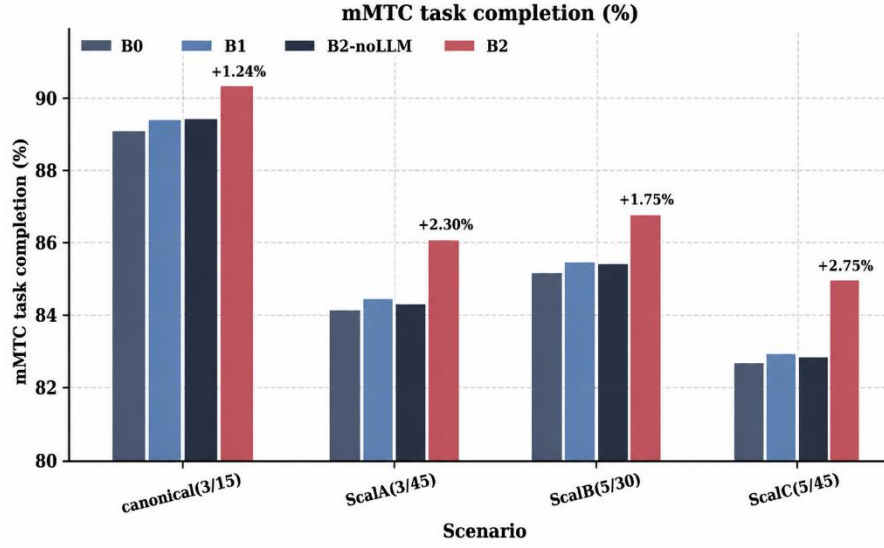


Fig. 5.4. mMTC Task Completion Rate Across Scenarios and Control Modes

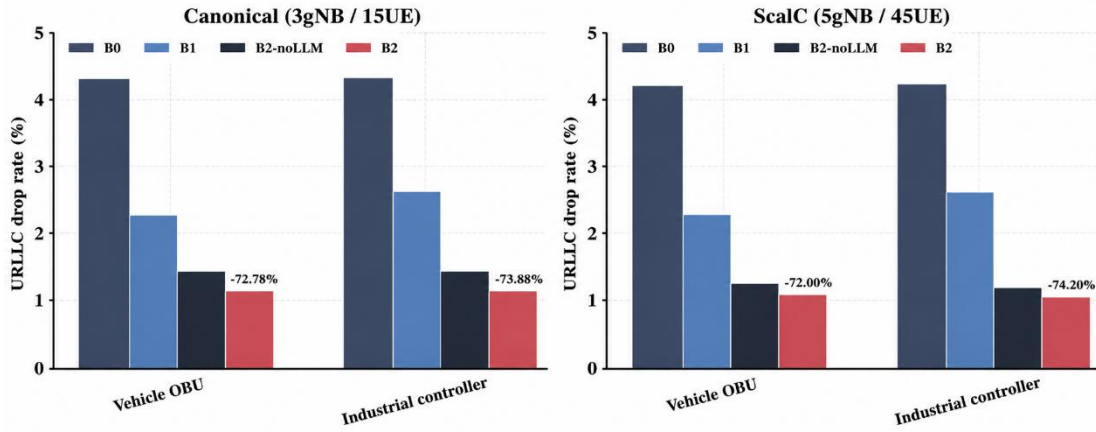


Fig. 5.5. Per-Device-Class URLLC Drop Rate Comparison

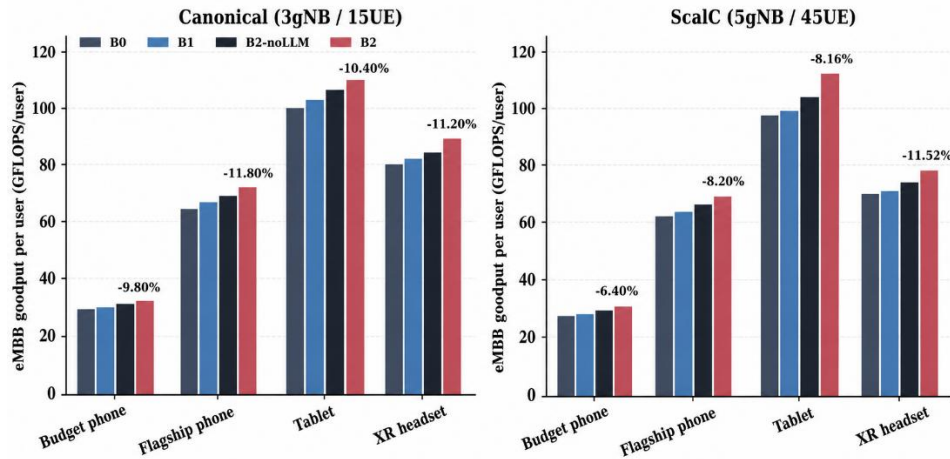


Fig. 5.6. Per-Device-Class eMBB Goodput Comparison

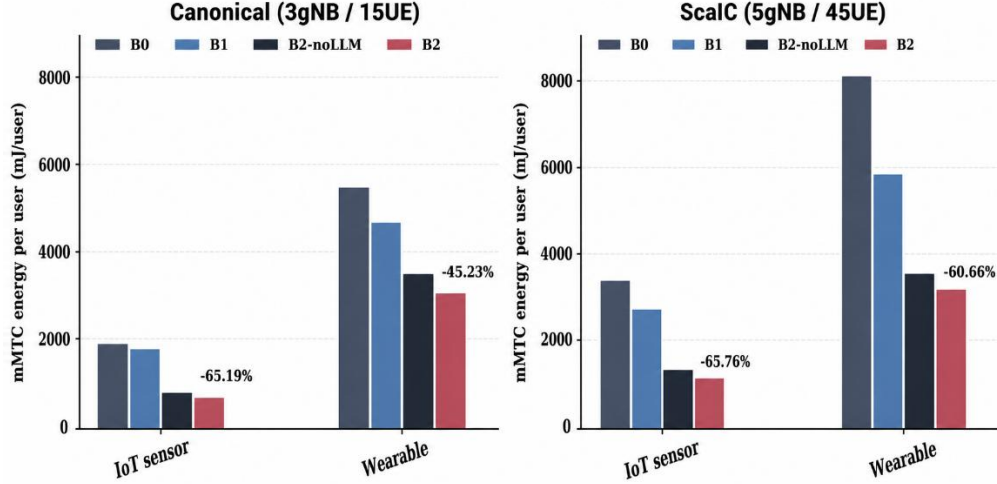


Fig. 5.7. Per-Device-Class mMTC Energy Consumption Comparison

The mechanism behind the energy savings: the 57% mMTC energy reduction between B0 and B2-TeleLLM is primarily driven by two components of the energy model. Reducing unnecessary handovers from 57.65 to 17.17 directly suppresses both the migration energy penalty (0.5 mJ per migration) and, more significantly, the radio re-establishment sequences that follow handovers, during which mMTC devices must re-acquire synchronization and transmit at higher power. Additionally, TeleLLM selects SAVE_MMTC_ENERGY eleven times per run versus three times for GPT-4o-mini in the canonical scenario. This bundle promotes local MEC placement for mMTC tasks and discourages migrations, with an effect that persists across multiple 60-second strategic windows.

The mechanism behind the goodput gains: the 18+ GFLOPS/s improvement over B2-noLLM results from approximately seven to eight LLM calls per 600-second run. When the optimizer selects FAVOR_EMBB or CONGESTION_RELIEF, admission priorities shift so that eMBB tasks receive priority access to local MEC resources and load redistribution releases compute capacity on congested nodes. These priority shifts persist for 60–120 seconds, during which hundreds of eMBB tasks benefit from better placement. PREEMPTIVE_REBALANCE also contributes by transferring tasks across the Xn mesh before saturation occurs, avoiding the queueing delays that most severely limit eMBB compute throughput.

5.9 Strategic Optimizer Behavior

Table 5.6 presents instrumentation data from the strategic optimizer in the canonical scenario, showing LLM call rates, shield intervention rates, and bundle preferences.

Table 5.6: Strategic Optimizer Instrumentation

Backend	Triggered calls	Skipped	Bundle switches	Rollbacks	Rejected	Top executed bundles
GPT-4o-mini	7.62	2.38	3.75	1.75	1.38	MOBILITY_STABILIZE=36, PREEMPTIVE_REBALANCE=11
TeleLLM	7.25	2.75	3.38	2.00	2.50	MOBILITY_STABILIZE=31, SAVE_MMTC_ENERGY=11

LLM call rates: 7.25–7.62 calls are triggered per run out of ten possible 60-second intervals, meaning 24–28% of intervals are skipped because the trigger conditions find no strategic value in changing the active bundle. This confirms that strategic decision-making is sparse and that the bulk of network management work is performed by the deterministic layers.

Safety shield activity: combining rollbacks and rejections as a fraction of triggered calls gives a modification rate of approximately 41% for GPT-4o-mini and 62% for TeleLLM in the canonical scenario. This is a central finding. The safety shield is not decorative; it actively modifies a substantial fraction of LLM decisions before they reach the network. The higher intervention rate for TeleLLM reflects its tendency to select more aggressive energy-saving and stability bundles that clash more frequently with the shield's hard constraints. Despite this higher intervention rate, both B2 modes consistently outperform B2-noLLM, demonstrating that the selections that pass through validation, combined with the corrections applied to those that do not, create sufficient net value.

Bundle preferences: TeleLLM selects SAVE_MMTC_ENERGY eleven times per run versus three for GPT-4o-mini in the canonical scenario. The frequency of MOBILITY_STABILIZE

is similar across backends (31 vs. 36). Notably, the higher MOBILITY_STABILIZE frequency for GPT-4o-mini does not produce a URLLC advantage, suggesting that TeleLLM's strength lies not primarily in which bundles it prefers overall, but in the precision of its contextual judgment about when each bundle is appropriate. This is exactly the type of context-sensitive decision-making that telecom-domain pretraining would strengthen.

Scalability of trigger logic: triggered call counts remain stable at 7.25–7.62 per run despite a 3× increase in UE count (15 to 45) and topology expansion (3 to 5 gNBs). This indicates that the trigger mechanism correctly identifies strategic decision points regardless of network size, which is encouraging for deployment scalability.

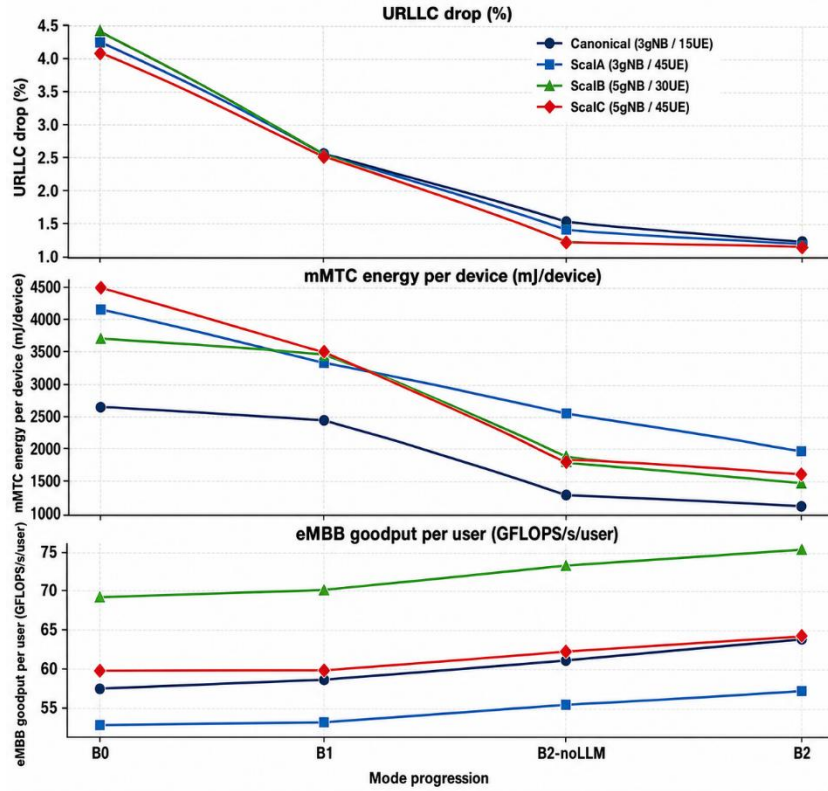


Fig. 5.8. Mode Progression of Key KPIs Across Scenarios

5.10 End-to-End Scalability

The most extreme configuration — 5 gNB / 45 UE joint scaling — tests the system when all control layers are simultaneously stressed.

Table 5.7: Scalability C: 5 gNB / 45 UE — Key Metrics (4-Seed Average)

Mode	URLLC drop %	eMBB goodput GFLOPS/s	mMTC energy mJ	Unnec. HO
B0	4.236	1,136.35	76,252	590.2
B1	2.457	1,139.19	58,804	331.5
B2-noLLM	1.235	1,190.14	32,371	213.9
B2- TeleLLM	1.148	1,219.99	27,764	176.5

The progressive improvement structure is fully preserved at scale. URLLC drop falls from 4.236% (B0) to 1.148% (B2-TeleLLM), a 73% reduction. mMTC energy decreases from 76,252 to 27,764 mJ (−64%), and unnecessary handovers fall from 590 to 177 (−70%). The eMBB goodput gain from B2-noLLM to B2-TeleLLM is 29.85 GFLOPS/s, the largest of any scenario, which is consistent with preemptive rebalancing being most effective under heavy concurrent demand.

A notable scaling observation is that the relative contribution of the LLM layer to mMTC energy reduction grows with scenario complexity: from 866 mJ in the canonical scenario (6.5% of the deterministic layers' contribution) to 4,607 mJ in the 5/45 scenario (12.5%). This indicates that the strategic LLM optimizer becomes more valuable precisely when the deterministic heuristics become strained — which is the desirable behavior for a slow strategic optimizer designed to resolve local optima that the fast deterministic loops cannot escape on their own.

5.11 Discussion: Domain Specialization and Safety Shield

The primary experimental finding is that TeleLLM, an 8-billion-parameter open-source model, outperforms GPT-4o-mini on URLLC drop, mMTC energy, and unnecessary handover prevention in all four scenarios while remaining competitive on eMBB metrics. Since the two models are comparable in architectural scale, domain specialization is the most plausible explanation for the performance difference, though it cannot be claimed as the sole variable given differences in instructional tuning strategy and inference infrastructure.

The task-specificity of bundle selection makes domain knowledge particularly relevant. The model does not need general reasoning ability; it needs to recognize that a state characterized by high mobility pressure and increasing URLLC fragility requires `MOBILITY_STABILIZE` rather than `CONGESTION_RELIEF`. This type of systematized telecom judgment is exactly what pretraining on 3GPP standards and IEEE publications would reinforce. The result extends prior work on telecom-domain LLMs [19], [14], [15], which report offline benchmark improvements from domain specialization; the present work generalizes that finding to closed-loop strategic control.

From a practical deployment perspective, a locally deployed telecom-fine-tuned model offers data sovereignty (no telemetry exits the operator's domain), vendor independence, lower inference cost per call, and, as demonstrated here, better performance on the metrics most critical for network reliability. The 15–16 second per-call inference latency on commodity GPU infrastructure is acceptable given the 60-second strategic interval, and would decrease to 2–5 seconds on specialized accelerator hardware.

Chapter 6

Conclusion, Limitations, and Future Work

6.1 Conclusion

This project proposed a hierarchical multi-agent system for joint handover and MEC optimization in 5G networks, incorporating a guarded LLM policy bundle selection mechanism at the slow strategic timescale. The system separates fast deterministic control from slow strategic reasoning, restricts LLM output to a finite library of eight pre-validated operating bundles, and enforces safety through bounded actuation, deterministic rejection, and rollback with anti-oscillation constraints.

The four-scenario, 20-run-per-mode benchmark yields four principal conclusions.

- 1- the deterministic substrate performs the majority of the work. The largest share of KPI improvement over the A3 baseline comes from intent-aware tower agents and the cross-layer deterministic supervisor. B1 reduces URLLC drop by 1.78–1.94 pp across all scenarios, while B2-noLLM adds another 1.02–1.27 pp and improves eMBB goodput by 17.8–50.9 GFLOPS/s. These deterministic layers collectively account for 91% of the total 73% URLLC drop reduction observed in the canonical scenario. This calibrates expectations for the LLM contribution and establishes that any further improvement to the LLM must be measured against a strong deterministic baseline.
- 2- guarded LLM bundle selection consistently adds value. In all four scenarios, both LLM backends improve eMBB completion, goodput, and mMTC energy relative to B2-noLLM. The safety shield modifies or reverses 40–62% of triggered LLM selections, yet the net effect remains positive in every case. Even constrained strategic guidance produces measurable and consistent gains.
- 3- domain specialization matters in closed-loop control. On architecturally comparable models, TeleLLM leads on URLLC drop in 4/4 scenarios and mMTC energy in 4/4 scenarios, while GPT-4o-mini leads on eMBB goodput in 4/4 scenarios. The performance differential is trade-off stable: domain-specialized

models advantage reliability and energy metrics, while general-purpose models advantage throughput metrics.

- 4- the safety shield is operationally essential, not cosmetic. The fact that 40–62% of LLM decisions are modified before reaching the network, and yet the system still consistently outperforms the deterministic baseline, demonstrates that the bounded-advisor pattern is both safe and effective. Any system deploying an LLM without such a shield would expose the network to the full set of suboptimal and potentially unsafe selections that the shield currently absorbs.

The overarching architectural conclusion is that the appropriate role for LLMs in 5G/6G network optimization is not to replace deterministic operations but to select operating regimes from a bounded, pre-validated library while deterministic agents enforce safety and execute control. The bundle selection boundary, which may appear to be a simple implementation detail, is in fact the mechanism that makes the entire integration safe, evaluable, and practically deployable. The evidence supports the feasibility of this pattern in simulation; physical testbed validation is the necessary next step before considering integration into production O-RAN intelligent controller architectures.

6.2 Limitations

Several limitations of this study should be acknowledged.

- 1- Statistical power: the 8-seed canonical scenario provides sufficient statistical power for the primary comparisons ($p < 0.01$ on key metrics). The 4-seed scalability scenarios establish directional consistency but lack the power to formally quantify small inter-backend differences or corner-case behavior.
- 2- Sampling conditions: both backends operate with temperature 0.7 and no top-p filtering, which controls sampling conditions for the comparison but does not establish that these are the optimal settings for bundle selection. Lower temperatures would reduce sampling variance and potentially increase context-appropriate selection rates; higher temperatures would increase exploration but also increase shield intervention rates.

- 3- Inference latency: TeleLLM's 15–16 second per-call inference latency on commodity GPU is acceptable for the 60-second strategic interval but would not be adequate if a much shorter strategic cycle were required. Specialized inference hardware would resolve this.
- 4- Single-operator scope: the architecture assumes centralized single-operator control. Extension to multi-operator scenarios involving competitive boundaries and shared spectrum coordination would require significant architectural additions and introduces game-theoretic complications not addressed in this work.
- 5- Simulation accuracy: the digital twin uses 3GPP radio models but does not simulate all aspects of a live network, including hardware faults, commercial scheduling policies, operator-specific priority rules, and real-world propagation effects beyond the UMa path-loss model. The Shannon-style throughput estimation is standard in simulation research but means that per-backend performance margins may differ in a real deployment with hardware scheduling and actual propagation variability. The findings support the architectural approach and its qualitative conclusions, but physical testbed validation is required before claims of production deployability can be made.
- 6- Wearable device nuance: the existing bundle library does not differentiate between mMTC device subtypes (wearables versus IoT sensors). Bundles that redistribute MEC load across the Xn mesh benefit URLLC and eMBB devices but disproportionately disrupt wearable locality, since small wearable tasks accumulate disproportionate migration energy relative to their compute demand. This is a recognized gap in the bundle design.

6.3 Future Work

Several directions for future investigation are evident from the findings of this project.

Bundle library automation: the current bundle library is designed manually. Constraint-aware optimization could expand the bundle action space by identifying Pareto-efficient parameter combinations and formally verifying them for safety, potentially improving the expressiveness of the library without sacrificing the bounded-action guarantee. A key

challenge is managing the trade-off between library size and LLM selection accuracy: more bundles may impair the model's ability to discriminate between similar alternatives.

mMTC device class disaggregation: future work should disaggregate mMTC device types at the bundle design level, potentially introducing device-class-aware energy postures or separate sub-bundles for wearables versus IoT sensors, addressing the locality disruption identified in Section 5.8.

Extended temporal horizons and online adaptation: future studies could evaluate the system over simulations spanning hours or days with persistent memory across sessions, testing whether the optimizer can learn daily traffic patterns without retraining and whether the experience buffer provides long-term adaptation benefits.

Multi-operator and federated scenarios: extending the architecture to multi-operator settings introduces both technical and game-theoretic challenges, including coordinating on shared spectrum while respecting competitive boundaries between operators. This represents a significant research direction for future 5G/6G network management.

Formal safety verification: the empirical safety record in this project is strong — no URLLC drop regression above 0.014 pp in any scenario — but formal verification methods, such as model checking of bundle parameter invariants and contract-based shield design, would provide provable safety guarantees that telecommunications regulators may require for critical infrastructure deployment.

Physical testbed validation: the most important next step is to validate the architecture on a physical 5G testbed, such as an O-RAN-compatible small-cell deployment, to confirm that the performance advantages observed in simulation translate to real hardware with its associated scheduling, propagation, and interference characteristics.

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الخلاصة

في شبكات الجيل الخامس الحديثة، يتطلب التحكم المشترك بقرارات التسليم (Handover) وتفرغ الحوسبة الطرفية المتنقلة (MEC Offloading) تنسيقاً عبر مقاييس زمنية متعددة وأهداف متنافسة لا تستطيع الخوارزميات التفاعلية المعتمدة على القواعد التعامل معها بصورة كافية. يقترح هذا المشروع نظاماً هرمياً متعدد الوكلاء يقسم التحكم الشبكي وفقاً للمقياس الزمني. على المستوى الأدنى، تمتلك كل محطة gNB وكيل برج محلي يتخذ قرارات سريعة خاصة بالتسليم وتخصيص MEC باستخدام متجه نية المستخدم الحالي (UE Intent Vector). يقوم مشرف حتمي متعدد الطبقات (Deterministic Cross-Layer Supervisor) بحل التعارضات التي تتطلب رؤية عبر عدة أبراج، بينما يعمل مُحسّن استراتيجي كل 60 ثانية لاختيار واحدة من ثماني حزم سياسات تم التحقق منها مسبقاً باستخدام نموذج لغوي كبير (LLM). يعمل الـ LLM فقط كمستشار مقيد، في حين يقوم درع أمان حتمي بفرض القيود الصارمة، ورفض الخيارات غير الآمنة، والتراجع عن التغييرات الضارة. تم تقييم النظام داخل توأم رقمي مستوحى من معايير 3GPP عبر أربعة سيناريوهات (5-3 محطات gNB، و15-45 جهاز UE). يأخذ التقييم بعين الاعتبار ثلاث فئات خدمات للجيل الخامس ذات متطلبات متنافسة: حيث تعطي حركة مرور URLLC الأولوية للكمون المنخفض جداً والاعتمادية، بينما تركز حركة مرور eMBB على الإنتاجية الحاسوبية العالية وإنجاز المهام، في حين تركز حركة مرور mMTC على كفاءة الطاقة واستقرار الاتصال واسع النطاق. تُظهر دراسة الاستئصال (Ablation Study) التي تتدرج من خط أساس 3GPP A3 (B0)، مروراً بوكلاء الأبراج المدركين للنية (B1)، ثم المشرف الحتمي (B2-noLLM)، وصولاً إلى اختيار الحزم المحمي باستخدام LLM — (B2) أن الطبقات الحتمية توفر معظم مكاسب الأداء.

يخفض B1 معدل إسقاط URLLC بمقدار 1.78-1.94 نقطة مئوية مقارنةً بـ B0، بينما يضيف B2-noLLM انخفاضاً إضافياً بمقدار 1.02-1.27 نقطة مئوية إلى جانب تحسينات كبيرة في إنتاجية eMBB. بعد ذلك، توفر طبقة LLM مكاسب إضافية متسقة عبر جميع السيناريوهات التي تم تقييمها. تم تقييم منصتين متقاربتين: GPT-4o-mini و TeleLLM. (LLaMA-3-8B-Tele-it) يحقق TeleLLM أفضل أداء في معدل إسقاط URLLC وكفاءة طاقة mMTC، بينما يتفوق GPT-4o-mini في إنتاجية eMBB. يقوم درع الأمان بتعديل أو عكس 40-62 بالمئة من اختيارات LLM، ومع ذلك فإن كلا وضعي B2 يتفوقان باستمرار على خط الأساس الحتمي. وعبر دراسة الاستئصال الكاملة، يقلل النظام الكامل ثلاثي الطبقات معدل إسقاط URLLC بنسبة 73 بالمئة مقارنةً بخط الأساس A3.



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بإشراف

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